

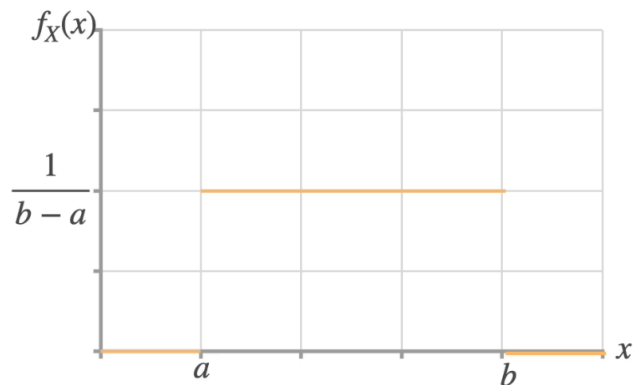
Uniform Distribution

A continuous random variable can be modeled with a **uniform** distribution if all possible values lie in an interval and have the **same frequency** of occurrence

Parameters:

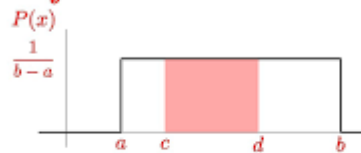
- a : beginning of the interval
- b : end of the interval

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & x \notin (a, b) \end{cases}$$



Uniform distribution is a probability distribution where every possible outcome within a specific range is equally likely to occur.

Uniform Distribution



Mean: $\mu = \frac{a+b}{2}$ Probability

S.D.: $\sigma = \sqrt{\frac{(b-a)^2}{12}}$ $P(c \leq X \leq d) = \frac{d-c}{b-a}$

